

Linear Algebra with Applications

lecture 08 Understanding Linear Transformations

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Introduction

This lecture explores the core concepts of linear transformations in vector spaces. We will learn how to represent any linear transformation as a matrix, which allows us to use the powerful tools of matrix algebra to analyze and apply these transformations. We will cover:

1. Finding the **standard matrix** of a linear transformation.
2. Deriving a transformation matrix by inspecting its output.
3. Understanding **composite transformations** like rotations and reflections.
4. Analyzing key properties: determining if a transformation is **one-to-one** or **onto**.

1 Finding the Standard Matrix of a Linear Transformation

💡 Key Idea

A linear transformation T that maps vectors from $\mathbb{R}^n$ to $\mathbb{R}^m$ can always be represented by a unique $m \times n$ matrix A . This matrix is called the **standard matrix** for the linear transformation T .

The key to finding this matrix is to observe how the transformation T acts on the **standard basis vectors** of the input space $\mathbb{R}^n$. The standard basis vectors, denoted as $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$, are vectors with a 1 in one position and zeros everywhere else.

The columns of the standard matrix A are simply the transformed basis vectors:

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \cdots \quad T(\mathbf{e}_n)]$$

Once we have the matrix A , applying the transformation T to any vector $\mathbf{x}$ is equivalent to the matrix-vector product $A\mathbf{x}$.

$$T(\mathbf{x}) = A\mathbf{x}$$

1.1 Example 1: Finding A from Transformed Basis Vectors

Problem: A linear transformation T maps vectors from $\mathbb{R}^3$ to $\mathbb{R}^4$. We are given the following information:

$$\bullet T(\mathbf{e}_1) = \begin{bmatrix} 3 \\ 2 \\ -1 \\ 0 \end{bmatrix}$$

$$\bullet T(\mathbf{e}_2) = \begin{bmatrix} 9 \\ 4 \\ 5 \\ 0 \end{bmatrix}$$

$$\bullet T(\mathbf{e}_3) = \begin{bmatrix} -1 \\ 0 \\ 0 \\ -2 \end{bmatrix}$$

Where $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ are the standard basis vectors in $\mathbb{R}^3$:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Tasks:

- Find the standard matrix A for the linear transformation T .
- Compute $T(\mathbf{x})$ for the vector $\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$.

Solution (Step-by-Step)

a) Find the Standard Matrix A

- Recall the Definition:** The standard matrix A is formed by using the transformed basis vectors $T(\mathbf{e}_1), T(\mathbf{e}_2)$, and $T(\mathbf{e}_3)$ as its columns.
- Construct the Matrix:** We place the given output vectors into the columns of our matrix A .

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad T(\mathbf{e}_3)] = \begin{bmatrix} 3 & 9 & -1 \\ 2 & 4 & 0 \\ -1 & 5 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

This is the standard matrix. Note that A is a 4×3 matrix, which is correct for a transformation from $\mathbb{R}^3$ to $\mathbb{R}^4$.

b) Compute $T(\mathbf{x})$ for $\mathbf{x} = [2, -1, 3]$

- Express $\mathbf{x}$ as a Linear Combination of Basis Vectors:**

$$\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - 1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 2\mathbf{e}_1 - \mathbf{e}_2 + 3\mathbf{e}_3$$

- Apply the Transformation T and Use Linearity:**

$$\begin{aligned} T(\mathbf{x}) &= T(2\mathbf{e}_1 - \mathbf{e}_2 + 3\mathbf{e}_3) \\ &= 2T(\mathbf{e}_1) - T(\mathbf{e}_2) + 3T(\mathbf{e}_3) \quad (\text{by linearity}) \end{aligned}$$

- Substitute the Known Values:**

$$T(\mathbf{x}) = 2 \begin{bmatrix} 3 \\ 2 \\ -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 9 \\ 4 \\ 5 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 0 \\ 0 \\ -2 \end{bmatrix}$$

- Perform Scalar Multiplication and Vector Addition:**

$$\begin{aligned} T(\mathbf{x}) &= \begin{bmatrix} 6 \\ 4 \\ -2 \\ 0 \end{bmatrix} - \begin{bmatrix} 9 \\ 4 \\ 5 \\ 0 \end{bmatrix} + \begin{bmatrix} -3 \\ 0 \\ 0 \\ -6 \end{bmatrix} \\ &= \begin{bmatrix} 6 - 9 - 3 \\ 4 - 4 + 0 \\ -2 - 5 + 0 \\ 0 - 0 - 6 \end{bmatrix} = \begin{bmatrix} -6 \\ 0 \\ -7 \\ -6 \end{bmatrix} \end{aligned}$$

Note: the video transcript calculation had a small error. '6-9-3 = -6'.

2 Deriving a Transformation Matrix by Inspection

💡 Key Idea

The matrix-vector product $A\mathbf{x}$ can be interpreted as a **linear combination of the columns of A**, where the weights are the components of the vector $\mathbf{x}$. If

$$A = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n] \text{ and } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \text{ then:}$$

$$A\mathbf{x} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_n\mathbf{v}_n$$

We can use this idea in reverse to find the columns of A if we know the general form of the output vector.

2.1 Example 2: Finding A by Inspection

Problem: A linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by $T(\mathbf{x}) = A\mathbf{x}$. The output is:

$$T(\mathbf{x}) = \begin{bmatrix} 3x_1 - x_3 \\ -2x_1 + 4x_2 \\ 6x_2 - x_3 \end{bmatrix}$$

Find the matrix A .

Solution (Step-by-Step)

1. **Separate the Output by Components of $\mathbf{x}$:** We rewrite the output by grouping terms associated with x_1, x_2 , and x_3 .

$$T(\mathbf{x}) = \begin{bmatrix} 3x_1 \\ -2x_1 \\ 0x_1 \end{bmatrix} + \begin{bmatrix} 0x_2 \\ 4x_2 \\ 6x_2 \end{bmatrix} + \begin{bmatrix} -1x_3 \\ 0x_3 \\ -1x_3 \end{bmatrix}$$

2. **Factor out the Scalars x_1, x_2, x_3 :**

$$T(\mathbf{x}) = x_1 \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 4 \\ 6 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix}$$

3. **Identify the Columns of A:** By comparing this to the definition $A\mathbf{x} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + x_3\mathbf{v}_3$, we can directly identify the columns of A :

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 4 \\ 6 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix}$$

4. **Construct the Matrix A:**

$$A = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] = \begin{bmatrix} 3 & 0 & -1 \\ -2 & 4 & 0 \\ 0 & 6 & -1 \end{bmatrix}$$

3 Composite Linear Transformations: Rotation and Reflection

💡 Key Idea

A complex linear transformation can often be described as a sequence of simpler transformations. If a transformation T consists of applying T_1 first, and then T_2 , the overall transformation matrix A is the product of the individual matrices A_1 and A_2 .

Crucial Point on Order: The order of matrix multiplication is critical. If T_1 is applied first, its matrix A_1 must be on the **right**.

$$T(\mathbf{x}) = T_2(T_1(\mathbf{x})) = A_2(A_1\mathbf{x}) = (A_2A_1)\mathbf{x}$$

So, the composite matrix is $A = A_2A_1$.

3.1 Example 3: Rotation followed by Reflection

Problem: A linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ first rotates points clockwise by $\pi/6$ radians about the origin, and then reflects them through the line $x_1 = -x_2$.

Tasks:

- Find the standard matrix A for this composite transformation.
- Find where the point $(1, 0)$ is mapped to by this transformation.

Solution (Step-by-Step)

a) Find the Standard Matrix A

- Find the Rotation Matrix (A_{rot}):** For a clockwise rotation by θ :

$$A_{rot} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$$

With $\theta = \pi/6$, we have $\cos(\pi/6) = \sqrt{3}/2$ and $\sin(\pi/6) = 1/2$.

$$A_{rot} = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}$$

- Find the Reflection Matrix (A_{ref}):** Reflection across $x_1 = -x_2$ sends a point (x_1, x_2) to $(-x_2, -x_1)$. The matrix is:

$$A_{ref} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

- Compose the Transformations:** The rotation happens first, then the reflection. So, $A = A_{ref}A_{rot}$.

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}$$

- Perform Matrix Multiplication:**

$$\begin{aligned} A &= \begin{bmatrix} (0)(\sqrt{3}/2) + (-1)(-1/2) & (0)(1/2) + (-1)(\sqrt{3}/2) \\ (-1)(\sqrt{3}/2) + (0)(-1/2) & (-1)(1/2) + (0)(\sqrt{3}/2) \end{bmatrix} \\ &= \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix} \end{aligned}$$

b) Apply the Transformation to the point (1, 0)

1. Set up the Matrix-Vector Product:

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

2. **Calculate the Result:** Multiplying by $\mathbf{e}_1 = [1, 0]$ selects the first column of the matrix.

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1/2 \\ -\sqrt{3}/2 \end{bmatrix}$$

4 Properties: One-to-One and Onto Transformations

4.1 One-to-One (Injective) Transformations

💡 Key Idea

Definition: A transformation T is **one-to-one** if every distinct input vector maps to a distinct output vector. That is, if $T(\mathbf{x}) = T(\mathbf{y})$, then it must be that $\mathbf{x} = \mathbf{y}$.

The Test: A transformation $T(\mathbf{x}) = A\mathbf{x}$ is one-to-one **if and only if the columns of A are linearly independent**.

How to Check: Solve the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

- If the **only** solution is the trivial solution ($\mathbf{x} = \mathbf{0}$), the columns are **linearly independent**, and T is **one-to-one**.
- If there are non-trivial solutions (indicated by a free variable), the columns are **linearly dependent**, and T is **not one-to-one**.

4.2 Onto (Surjective) Transformations

💡 Key Idea

Definition: A transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **onto** if its range covers the entire codomain. For any vector $\mathbf{b}$ in $\mathbb{R}^m$, there exists at least one input vector $\mathbf{x}$ in $\mathbb{R}^n$ such that $T(\mathbf{x}) = \mathbf{b}$.

The Test: A transformation $T(\mathbf{x}) = A\mathbf{x}$ is onto **if and only if the columns of A span the codomain $\mathbb{R}^m$** .

How to Check: For the columns of A to span $\mathbb{R}^m$, you must have a pivot position in every **row** of the matrix A (after row reduction).

4.3 Example 4: Checking for One-to-One and Onto

Problem: Determine if the transformations represented by the following matrices ($T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$) are one-to-one and onto.

a) $A = \begin{bmatrix} 1 & -3 & 4 \\ 3 & 0 & 0 \\ 2 & 1 & 1 \end{bmatrix}$

b) $B = \begin{bmatrix} 1 & -3 & 4 \\ 3 & 0 & 0 \\ 2 & 1 & 3 \end{bmatrix}$

Solution (Step-by-Step)

Since these are square matrices, the one-to-one and onto properties are linked. We only need to check for the linear independence of the columns.

a) Analysis of Matrix A

1. **Set up and Row Reduce:** To check for linear independence, we solve $A\mathbf{x} = \mathbf{0}$ by row reducing the augmented matrix $[A|\mathbf{0}]$.

$$\begin{bmatrix} 1 & -3 & 4 & 0 \\ 3 & 0 & 0 & 0 \\ 2 & 1 & 1 & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

2. **Analyze the Result:** There is a pivot in every column. This means there are no free variables, and the only solution is the trivial solution $\mathbf{x} = \mathbf{0}$.
3. **Conclusion:** The columns of A are **linearly independent**. Therefore, the transformation is **one-to-one** and **onto**.

b) Analysis of Matrix B

1. **Set up and Row Reduce:** We solve $B\mathbf{x} = \mathbf{0}$.

$$\begin{bmatrix} 1 & -3 & 4 & 0 \\ 3 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -4/3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2. **Analyze the Result:** The third column does not have a pivot. This means x_3 is a **free variable**, and there are infinitely many non-trivial solutions.
3. **Conclusion:** The columns of B are **linearly dependent**. Therefore, the transformation is **not one-to-one** and **not onto**.

5 Practical Problems in Machine Learning

This section explores how the concepts of linear transformations are directly applied in common Machine Learning tasks.

5.1 Feature Transformation in a Linear Model

Problem 1 In machine learning, we often perform **feature engineering** to create new, more informative features from existing ones. Suppose we have a 2D data point $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

and we create a new 3D feature vector $\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$ using the following linear combinations:

$$z_1 = x_1 - 2x_2$$

$$z_2 = 3x_1 + x_2$$

$$z_3 = 4x_2$$

This is a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$. Find the standard matrix A such that $\mathbf{z} = A\mathbf{x}$.

Solution

We can find the matrix A by inspection, as shown in Example 2. We express the output vector $\mathbf{z}$ as a linear combination of vectors weighted by x_1 and x_2 .

$$\begin{aligned}\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} &= \begin{bmatrix} x_1 - 2x_2 \\ 3x_1 + x_2 \\ 0x_1 + 4x_2 \end{bmatrix} \\ &= \begin{bmatrix} 1x_1 \\ 3x_1 \\ 0x_1 \end{bmatrix} + \begin{bmatrix} -2x_2 \\ 1x_2 \\ 4x_2 \end{bmatrix} \\ &= x_1 \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}\end{aligned}$$

From the form $\mathbf{z} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2$, the columns of our matrix A are the vectors multiplying x_1 and x_2 .

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 1 \\ 0 & 4 \end{bmatrix}$$

This 3×2 matrix performs the specified feature transformation.

5.2 Data Augmentation for Image Data

Problem 2 **Data augmentation** is a technique used to increase the size of a training dataset by creating modified versions of existing data. For a 2D image, a point (x, y) can be transformed. Consider an augmentation pipeline that first **flips an image horizontally** (reflection across the y-axis) and then **rotates it 90 degrees counter-clockwise**.

Find the single transformation matrix that performs this composite operation.

Solution

Let's find the matrix for each step.

1. **Horizontal Flip (T_{flip}):** This transformation sends a point $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ to $\begin{bmatrix} -x \\ y \end{bmatrix}$. The matrix is:

$$A_{flip} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

2. **Counter-Clockwise 90° Rotation (T_{rot}):** The rotation matrix for an angle θ is $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$. For $\theta = 90^\circ$, this becomes:

$$A_{rot} = \begin{bmatrix} \cos(90^\circ) & -\sin(90^\circ) \\ \sin(90^\circ) & \cos(90^\circ) \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

The flip is applied first, so A_{flip} is on the right. The final matrix A is:

$$A = A_{rot}A_{flip} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} (0)(-1) + (-1)(0) & (0)(0) + (-1)(1) \\ (1)(-1) + (0)(0) & (1)(0) + (0)(1) \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

This single matrix performs both operations.

5.3 Analyzing Dimensionality Reduction (PCA)

Problem 3 **Principal Component Analysis (PCA)** is a popular dimensionality reduction technique. It projects data from a high-dimensional space to a lower-dimensional space. Suppose a PCA model transforms data from $\mathbb{R}^3$ to $\mathbb{R}^2$ using the matrix A :

$$A = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} \end{bmatrix}$$

Analyze this transformation $T(\mathbf{x}) = A\mathbf{x}$. Is it one-to-one? Is it onto? What does this mean for the data?

Solution

- **Is it one-to-one?** A transformation is one-to-one if its columns are linearly independent. The matrix A is 2×3 . It has 3 column vectors in $\mathbb{R}^2$. A set of 3 vectors in a 2-dimensional space must be linearly dependent.

Alternatively, to solve $A\mathbf{x} = \mathbf{0}$, we see that the system has 3 variables and at most 2 pivots (since there are only 2 rows). This guarantees at least one free variable. Therefore, non-trivial solutions exist, the columns are linearly dependent, and the transformation is **not one-to-one**.

ML Interpretation: "Not one-to-one" means that multiple input vectors in $\mathbb{R}^3$ can map to the same output vector in $\mathbb{R}^2$. This is the definition of **information loss**, which is an inherent part of dimensionality reduction.

- **Is it onto?** A transformation is onto $\mathbb{R}^2$ if its columns span $\mathbb{R}^2$. This requires two linearly independent columns. Let's check the first two columns: $\mathbf{v}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{6} \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{6} \end{bmatrix}$. They are not scalar multiples of each other, so they are linearly independent and thus span $\mathbb{R}^2$.

Therefore, the transformation is **onto**.

ML Interpretation: "Onto" means that any point in the new 2D feature space can be produced by some point from the original 3D space. The projection covers the entire target space, which is a desirable property.

6 Summary and Exercise Problems

6.1 Summary of Key Concepts

- **Standard Matrix:** Any linear transformation T can be written as $T(\mathbf{x}) = A\mathbf{x}$, where $A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \cdots]$.
- **Composite Transformations:** A sequence of transformations corresponds to the product of their matrices, applied from right to left.
- **One-to-One:** Equivalent to having **linearly independent columns** in A .
- **Onto:** Equivalent to the **columns of A spanning the codomain**.
- **Square Matrices:** For an $n \times n$ matrix, being one-to-one, onto, and having linearly independent columns are all equivalent properties.

6.2 Exercise Problems

1. A linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is defined by $T(\mathbf{e}_1) = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$ and $T(\mathbf{e}_2) = \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}$.

Find the standard matrix A and compute $T(\mathbf{x})$ for $\mathbf{x} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$.

2. A linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ first reflects points across the y -axis and then rotates them counter-clockwise by 90° ($\pi/2$ radians). Find the single standard matrix A that represents this composite transformation. (Hint: Reflection across y -axis sends (x_1, x_2) to $(-x_1, x_2)$).

3. Consider the linear transformation $T(\mathbf{x}) = A\mathbf{x}$ where $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 2 \end{bmatrix}$. Determine if this transformation is one-to-one and onto. Justify your answer.

4. Can a linear transformation $T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ be one-to-one? Can it be onto? Explain your reasoning.

7 Summary and Applications

7.1 Summary of Key Concepts

- **Standard Matrix:** Any linear transformation T can be written as $T(\mathbf{x}) = A\mathbf{x}$, where $A = [T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ \cdots]$.
- **Composite Transformations:** A sequence of transformations corresponds to the product of their matrices, applied from right to left.
- **One-to-One:** Equivalent to having **linearly independent columns** in A . The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- **Onto:** Equivalent to the **columns of A spanning the codomain**. For an $m \times n$ matrix, this means there is a pivot in every row.
- **Square Matrices:** For an $n \times n$ matrix, being one-to-one, onto, and having linearly independent columns are all equivalent properties.

Practical Applications in Data Science & Machine Learning

- **Dimensionality Reduction:** Techniques like Principal Component Analysis (PCA) use linear transformations to project high-dimensional data onto a lower-dimensional space. This transformation is neither one-to-one (information is lost) nor onto.
- **Computer Graphics & Image Processing:** Rotations, scaling, and shearing are fundamental linear transformations used to manipulate images and 3D models.
- **Data Augmentation:** In training neural networks, new training data is created by applying random transformations (rotations, flips) to existing images.
- **Natural Language Processing (NLP):** Word embedding models represent words as vectors. Linear transformations can map these vectors between languages for machine translation or to analyze word relationships.

8 Exercise Problems

1. A linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is defined by $T(\mathbf{e}_1) = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$ and $T(\mathbf{e}_2) = \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}$.

Find the standard matrix A and compute $T(\mathbf{x})$ for $\mathbf{x} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$.

2. A linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ first reflects points across the y-axis and then rotates them counter-clockwise by 90° ($\pi/2$ radians). Find the single standard matrix A that represents this composite transformation. (Hint: Reflection across y-axis sends (x_1, x_2) to $(-x_1, x_2)$).

3. Consider the linear transformation $T(\mathbf{x}) = A\mathbf{x}$ where $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 2 \end{bmatrix}$. Determine if this transformation is one-to-one and onto. Justify your answer.

4. Can a linear transformation $T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ be one-to-one? Can it be onto? Explain your reasoning.